

CHAPTER 1

FOUNDATION OF DATA

Data -Meaning

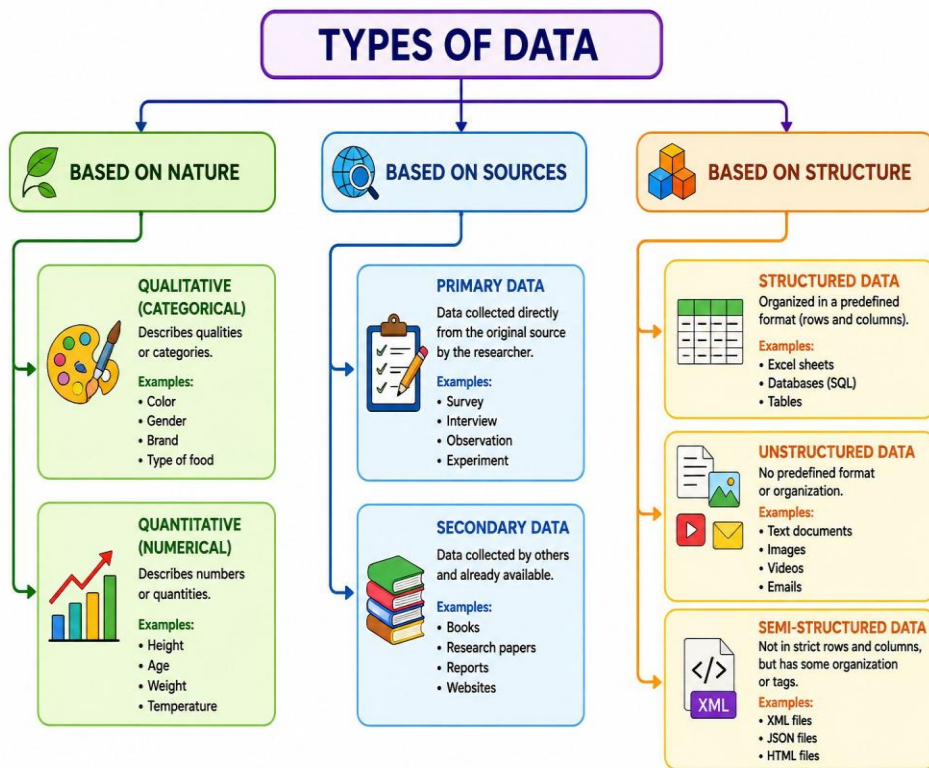
Data means facts, numbers, or information collected for reference or analysis.

Example: Websites, social media, Online shopping, Business analytics, E-Mail, Mobile devices

Data Surrounds Us



Types of Data



1. Based on Nature

a) Qualitative Data

b) Quantitative Data

2. Based on Source

a) Primary Data

b) Secondary Data

3. Based on Structure

a) Structured Data

b) Unstructured Data

c) Semi-Structured Data

1. Based on Nature

a) Qualitative Data (Categorical)

- *Nominal Data (no order)*

Data that is categorized without any order or ranking.

Example: Gender, Colors, Country, Blood group.

- *Ordinal (ordered)*

Data that has a clear order or ranking, but the gap between them is not defined.

Example: Education level, Class rank, Satisfaction level, Rating scale.

b) Quantitative Data (Numerical)

- *Discrete: Countable values*

Example: Number of students =25, Number of cars in a parking area = 25, Time and Date.

- *Continuous: Measurable values*

Example: Time, Speed, Amount of water, Weight, height, Temperature

2. Based on Source

a) Primary Data: Collected first hand

Example: Survey, Experiment, Observation, Feedback from customers

b) Secondary Data: Already available.

Example: Reports, Books, Research papers

3. Based on Structure:

a) Structured Data:

Data that highly organized in rows and columns.

Example: Excel sheet, Databases, Student mark lists

b) Unstructured Data:

Data that does not have a fixed format.

Example: Text, Images, Videos, Audio

c) Semi-Structured Data:

Data that has some structure but not fully organized.

Example: Emails, JSON / XML files, Web pages.

CHAPTER 2

DATA ANALYSIS

Data analysis is the process of collecting, cleaning, transforming and interpreting data to discover meaningful insights, support decision-making and predict future outcomes.

Objectives of Data Analytics

- *Identify trends and pattern*
- *Support Strategic and operational decisions.*
- *Convert raw data into actionable knowledge*

Daily life Applications

- 1. Online shopping*
- 2. Personal Finance*
- 3. Screen time analysis*
- 4. Instagram post analysis*

5.Fitness tracking

6.Weather forecast

7.Education

8.Traffic Management

9.Social Media insights

10.Hospital and Healthcare -work

11.Logistics and delivery

12.Restaurant business

13.Agriculture

14.Banking

1. Online Shopping

Helps websites suggest products based on customer interests and previous purchases.

2. Personal Finance

Tracks income, expenses, savings, and helps manage budgets better.

3. Screen Time Analysis

Monitors mobile or app usage to improve productivity and reduce overuse.

4. Instagram Post Analysis

Analyzes likes, comments, shares, and reach to understand audience engagement.

5. Fitness Tracking

Tracks steps, calories, heart rate, sleep, and workout performance.

6. Weather Forecast

Uses weather data to predict rain, temperature, and climate conditions.

7. Education

Helps students and teachers track performance, attendance, and learning progress.

8. Traffic Management

Analyzes road traffic data to reduce congestion and improve transportation.

9. Social Media Insights

Measures trends, user behavior, and content performance on social platforms.

10. Hospital and Healthcare Work

Helps monitor patient records, disease trends, and improves treatment planning.

11. Logistics and Delivery

Tracks shipments, delivery routes, and delivery time for faster service.

12. Restaurant Business

Analyzes customer preferences, sales, and food demand to improve service.

13. Agriculture

Helps farmers monitor crop growth, weather, soil quality, and irrigation.

14. Banking

Detects fraud, manages customer accounts, analyzes transactions, and improves security.